



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

Artificial and Computational Intelligence

AIMLCZG557

ACI Team



AIMLCZG557 – CS#10

Knowledge Representation using Logics

Agenda for CS #10



1. Recap of CS#09
2. Introduction to Predicate Logic
3. Predicate Logic resolution using Forward & Backward chaining
4. Exercises

Course Progress



- M1 Introduction to AI
- M2 Problem Solving Agent using Search
- M3 Game Playing
- M4 Knowledge Representation using Logics
- M5 Probabilistic Representation and Reasoning
- M6 Reasoning over time
- M7 Ethics in AI

First-Order Logic / Predicate Logic



- First-order logic is also known as Predicate logic or First-order predicate logic.
- Whereas propositional logic assumes the world contains **facts**,
- first-order logic (like natural language) assumes the world contains
 - **Objects**: people, houses, numbers, colors, baseball games, wars, ...
 - **Relations**: red, round, prime, brother of, bigger than, part of, comes between, ...
 - **Functions**: father of, best friend, one more than, plus, ...

Syntax of FOL: Basic Elements



- **Constants** KingJohn, 2, NUS,...
- **Predicates** Brother, >,...
- **Functions** Sqrt, LeftLegOf, ...
- **Variables** x, y, a, b,...
- **Connectives** $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$
- **Equality** =
- **Quantifiers** $\forall$, $\exists$
- **Sentences** Atom / complex sentences
- **Atom** True / False / AP (atomic proposition)
- **Complex sentence** (sentence) / connective sentence / $\neg$ sentence

Atomic Sentences



Atomic sentence = $\text{predicate}(\text{term}_1, \dots, \text{term}_n)$
or $\text{term}_1 = \text{term}_2$

Term = $\text{function}(\text{term}_1, \dots, \text{term}_n)$ or *constant* or *variable*

- E.g., $\text{Brother}(\text{KingJohn}, \text{RichardTheLionheart})$
> $(\text{Length}(\text{LeftLegOf}(\text{Richard})),$
 $\text{Length}(\text{LeftLegOf}(\text{KingJohn})))$

Complex Sentences



- Complex sentences are made from atomic sentences using connectives

$$\neg S, S_1 \wedge S_2, S_1 \vee S_2, S_1 \Rightarrow S_2, S_1 \Leftrightarrow S_2,$$

E.g. *Sibling(KingJohn, Richard) $\Rightarrow$ Sibling(Richard, KingJohn)*

$$>(1,2) \vee \leq (1,2)$$

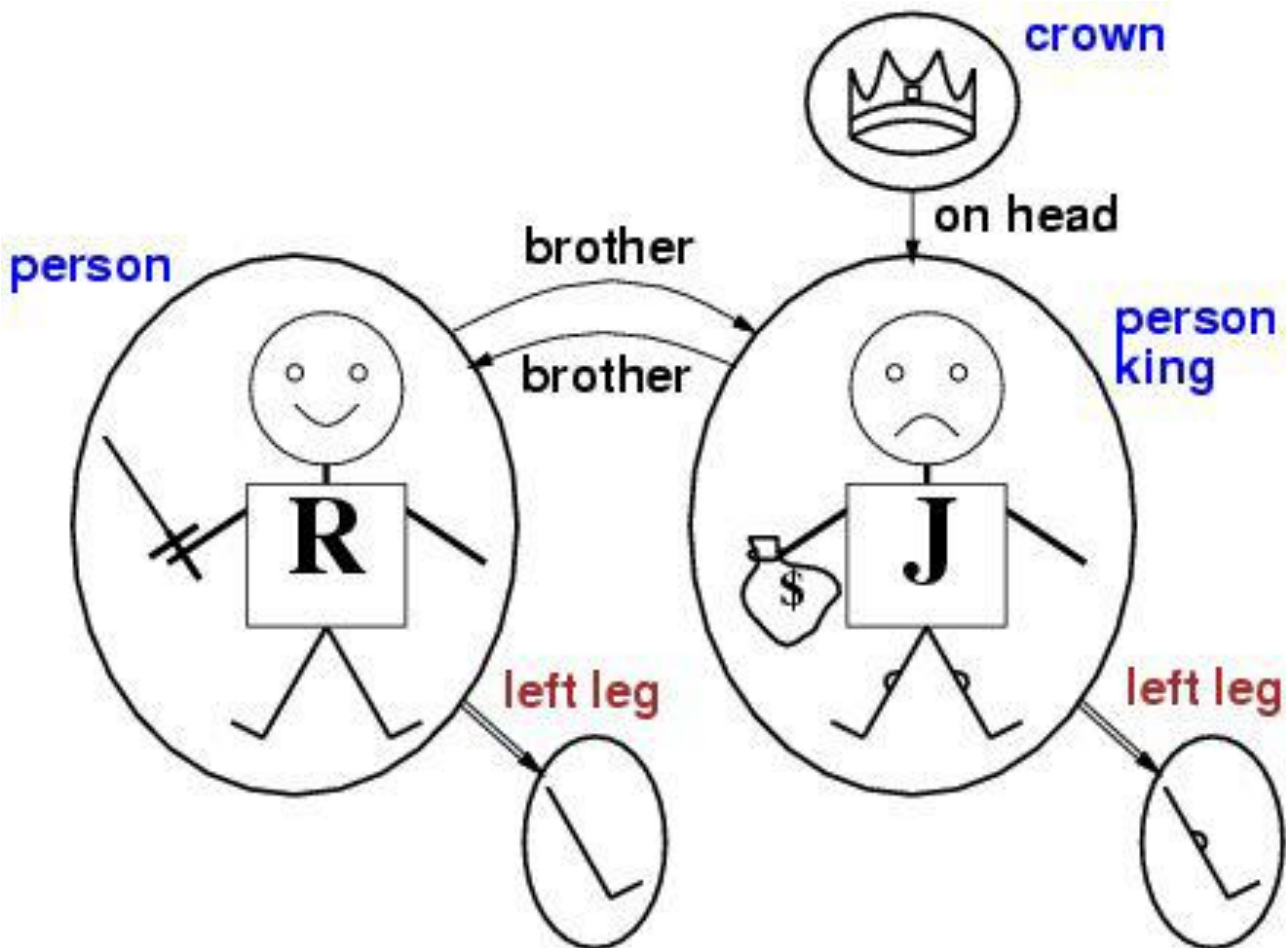
$$>(1,2) \wedge \neg >(1,2)$$

Truth in FOL



- Sentences are true with respect to a **model** and an **interpretation**
- Model contains objects (**domain elements**) and relations among them
- Interpretation specifies referents for
 - constant symbols** → **objects**
 - predicate symbols** → **relations**
 - function symbols** → **functional relations**
- An atomic sentence $predicate(term_1, \dots, term_n)$ is true iff the **objects** referred to by $term_1, \dots, term_n$ are in the **relation** referred to by $predicate$

Models of FOL: Example



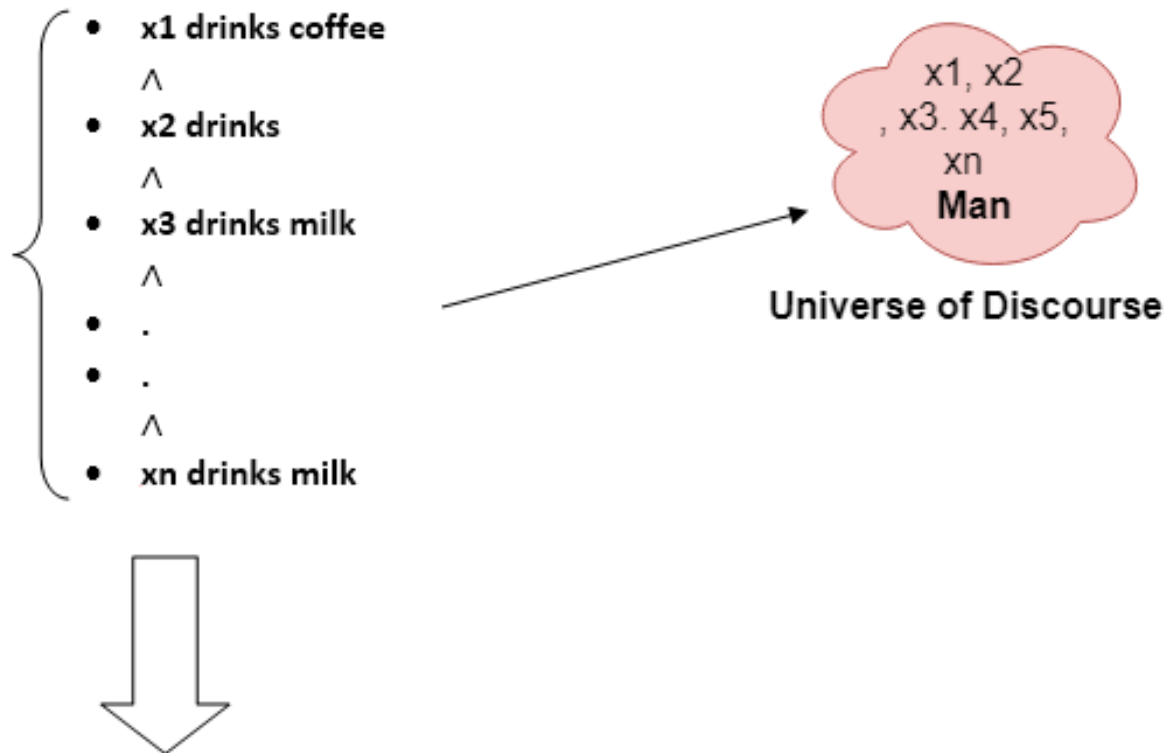
Universal Quantifier Examples

If x is a variable, then $\forall x$ is read as:

- **For all x**
- **For each x**
- **For every x .**

Example: **All man drink coffee.**

Universal Quantifier Examples



So in shorthand notation, we can write it as :

$\forall x \text{ man}(x) \rightarrow \text{drink}(x, \text{coffee}).$

It will be read as: There are all x where x is a man who drink coffee. 12

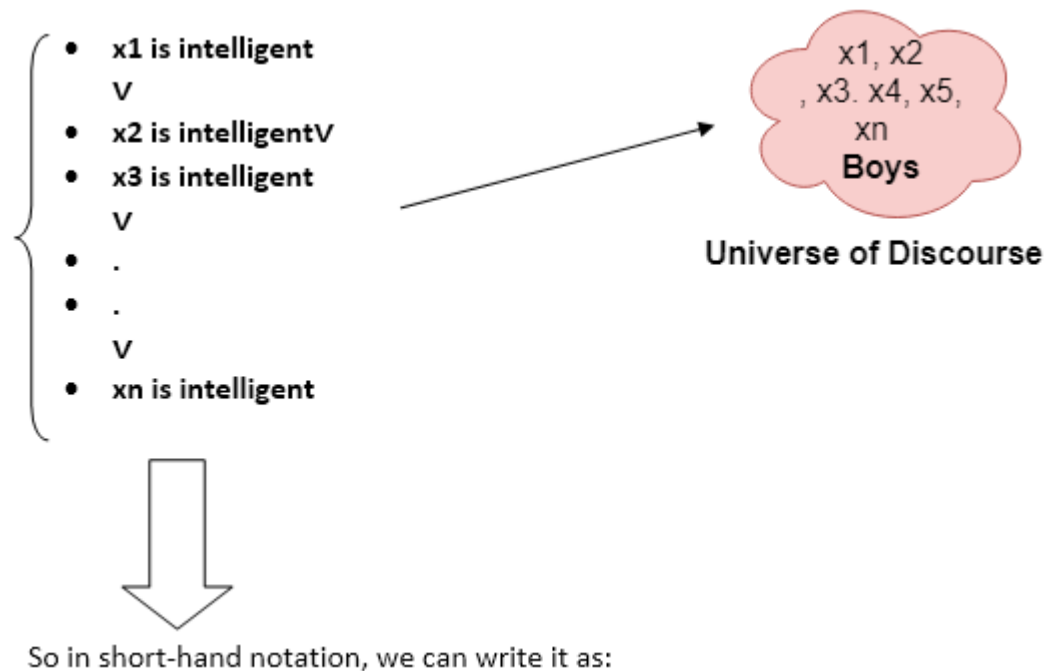
Existential Quantifier

If x is a variable, then existential quantifier will be $\exists x$ or $\exists(x)$. And it will be read as:

- **There exists a 'x.'**
- **For some 'x.'**
- **For at least one 'x.'**

Example: **Some boys are intelligent.**

Existential Quantifier



$\exists x: \text{boys}(x) \wedge \text{intelligent}(x)$

It will be read as: There are some x where x is a boy who is intelligent.

Examples

Prof John is teaching ML course for WILP Students

$\exists x, c, s$ Teaching (X, C, S) where X is Prof John, C is Course, S is WILP Students

Seller sells all the products to customers

$\forall p$ Sells (S, P, C)

Universal Quantification

- $\forall <variables> <sentence>$

Everyone at BITS is smart:

$$\forall x \text{ At}(x, \text{BITS}) \Rightarrow \text{Smart}(x)$$

- $\forall x P$ is true in a model m iff P is true with x being each possible object in the model
- Roughly speaking, equivalent to the conjunction of instantiations of P

$$\begin{aligned} & \text{At}(\text{KingJohn}, \text{BITS}) \Rightarrow \text{Smart}(\text{KingJohn}) \\ \wedge & \text{ At}(\text{Richard}, \text{BITS}) \Rightarrow \text{Smart}(\text{Richard}) \\ \wedge & \text{ At}(\text{BITS}, \text{BITS}) \Rightarrow \text{Smart}(\text{BITS}) \\ \wedge & \dots \end{aligned}$$

Existential Quantification

- $\exists \langle \text{variables} \rangle \langle \text{sentence} \rangle$
- Someone at NUS is smart:
- $\exists x \text{At}(x, \text{NUS}) \wedge \text{Smart}(x)$
- $\exists x P$ is true in a model m iff P is true with x being some possible object in the model
- Roughly speaking, equivalent to the **disjunction** of **instantiations** of P
 - $\text{At}(\text{KingJohn}, \text{NUS}) \wedge \text{Smart}(\text{KingJohn})$
 - $\vee \text{At}(\text{Richard}, \text{NUS}) \wedge \text{Smart}(\text{Richard})$
 - $\vee \text{At}(\text{NUS}, \text{NUS}) \wedge \text{Smart}(\text{NUS})$
 - $\vee \dots$

Properties of Quantifiers

- $\forall x \forall y$ is the same as $\forall y \forall x$
- $\exists x \exists y$ is the same as $\exists y \exists x$
- $\exists x \forall y$ is **not** the same as $\forall y \exists x$
- $\exists x \forall y \text{ Loves}(x,y)$
 - “There is a person who loves everyone in the world”
- $\forall y \exists x \text{ Loves}(x,y)$
 - “Everyone in the world is loved by at least one person”
- **Quantifier duality:** each can be expressed using the other
- $\forall x \text{ Likes}(x, \text{IceCream}) \quad \neg \exists x \neg \text{Likes}(x, \text{IceCream})$
- $\exists x \text{ Likes}(x, \text{Broccoli}) \quad \neg \forall x \neg \text{Likes}(x, \text{Broccoli})$

Universal Instantiation (UI)



- Every instantiation of a universally quantified sentence is entailed by it:

$$\frac{\forall v \alpha}{\text{Subst}(\{v/g\}, \alpha)}$$

for any variable v and ground term g

- E.g., $\forall x \text{ King}(x) \wedge \text{Greedy}(x) \Rightarrow \text{Evil}(x)$ yields:
 $\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \Rightarrow \text{Evil}(\text{John})$
 $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \Rightarrow \text{Evil}(\text{Richard})$
 $\text{King}(\text{Father}(\text{John})) \wedge \text{Greedy}(\text{Father}(\text{John})) \Rightarrow \text{Evil}(\text{Father}(\text{John}))$

Existential Instantiation (EI)



- For any sentence α , variable v , and constant symbol k that does **not** appear elsewhere in the knowledge base:

$$\frac{\exists v \alpha}{\text{Subst}(\{v/k\}, \alpha)}$$

- E.g., $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$ yields:

$$\text{Crown}(C_1) \wedge \text{OnHead}(C_1, \text{John})$$

provided C_1 is a new constant symbol, called a **Skolem constant**

Reduction to Propositional Inference



Suppose the KB contains just the following:

$\forall x \text{ King}(x) \wedge \text{Greedy}(x) \Rightarrow \text{Evil}(x)$

$\text{King}(\text{John})$

$\text{Greedy}(\text{John})$

$\text{Brother}(\text{Richard}, \text{John})$

- Instantiating the universal sentence in **all possible** ways, we have:

$\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \Rightarrow \text{Evil}(\text{John})$

$\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \Rightarrow \text{Evil}(\text{Richard})$

$\text{King}(\text{John})$

$\text{Greedy}(\text{John})$

$\text{Brother}(\text{Richard}, \text{John})$

- The new KB is **propositionalized**: proposition symbols are

$\text{King}(\text{John}), \text{Greedy}(\text{John}), \text{Evil}(\text{John}), \text{King}(\text{Richard}), \text{etc.}$

Exercises

Convert the following into FOL using quantifiers:

1. All birds fly.
2. Every man respects his parent.
3. Some boys play cricket.
4. Not all students like both Mathematics and Science.
5. Only one student failed in Mathematics.

Forward Chaining

- Consider the following problem:

The law says it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.

- We will prove that West is a criminal

Algorithm:

1. Start from the facts
2. Trigger all rules whose premises are satisfied
3. **Add the conclusion to known facts**
4. Repeat the steps till no new facts are added or the query is answered

Forward Chaining

- “All of its missiles were sold to it by Colonel West”

$$Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$$

- Missiles are weapons

$$Missile(x) \Rightarrow Weapon(x)$$

- Hostile means enemy

$$Enemy(x, America) \Rightarrow Hostile(x)$$

- “West, who is American”

$$American(West)$$

- “The country Nono, an enemy of America”

$$Enemy(Nono, America)$$

Forward Chaining

- First, we will represent the facts in First Order Definite Clauses

“ ... it is a crime for an American to sell weapons to hostile nations”

$$American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$$

“Nono ... has some missiles”

$$\exists x Owns(Nono, x) \wedge Missile(x)$$

is transformed into two definite clauses by Existential Instantiation

$$Owns(Nono, M_1)$$

$$Missile(M_1)$$

Forward Chaining

- ① $American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$
- ② $Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$
- ③ $Missile(x) \Rightarrow Weapon(x)$
- ④ $Enemy(x, America) \Rightarrow Hostile(x)$

Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)

Forward Chaining

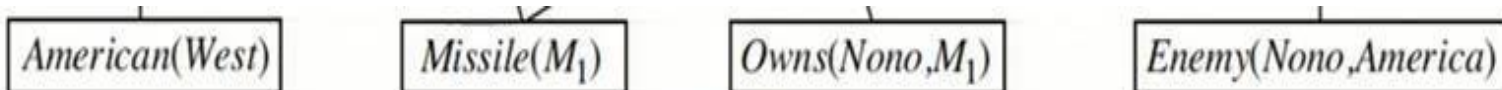
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- ② $Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$
- ③ $Missile(x) \Rightarrow Weapon(x)$
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Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Forward Chaining

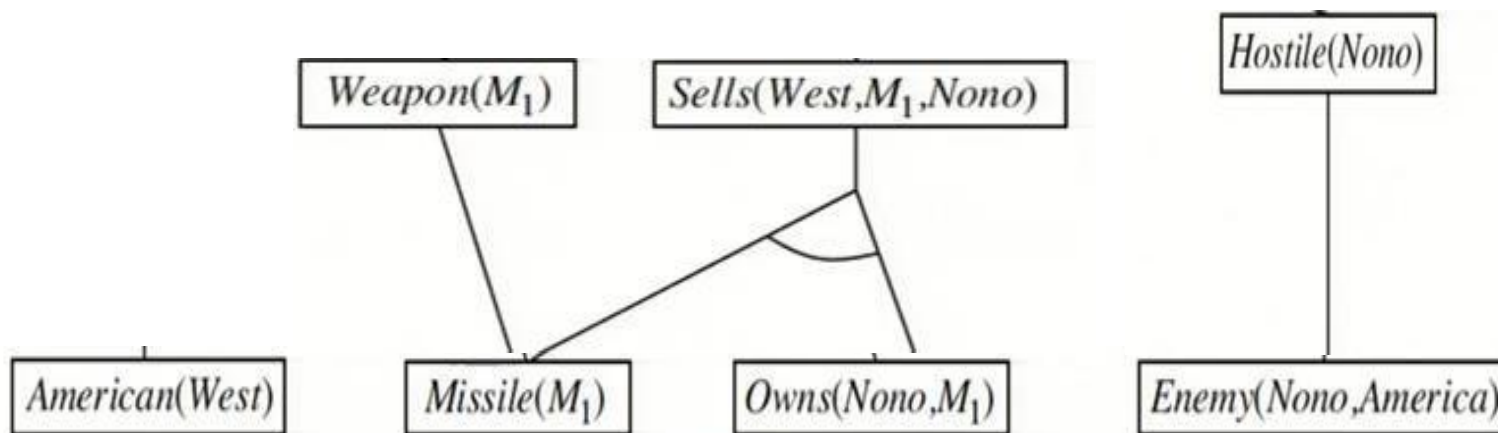
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Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Forward Chaining

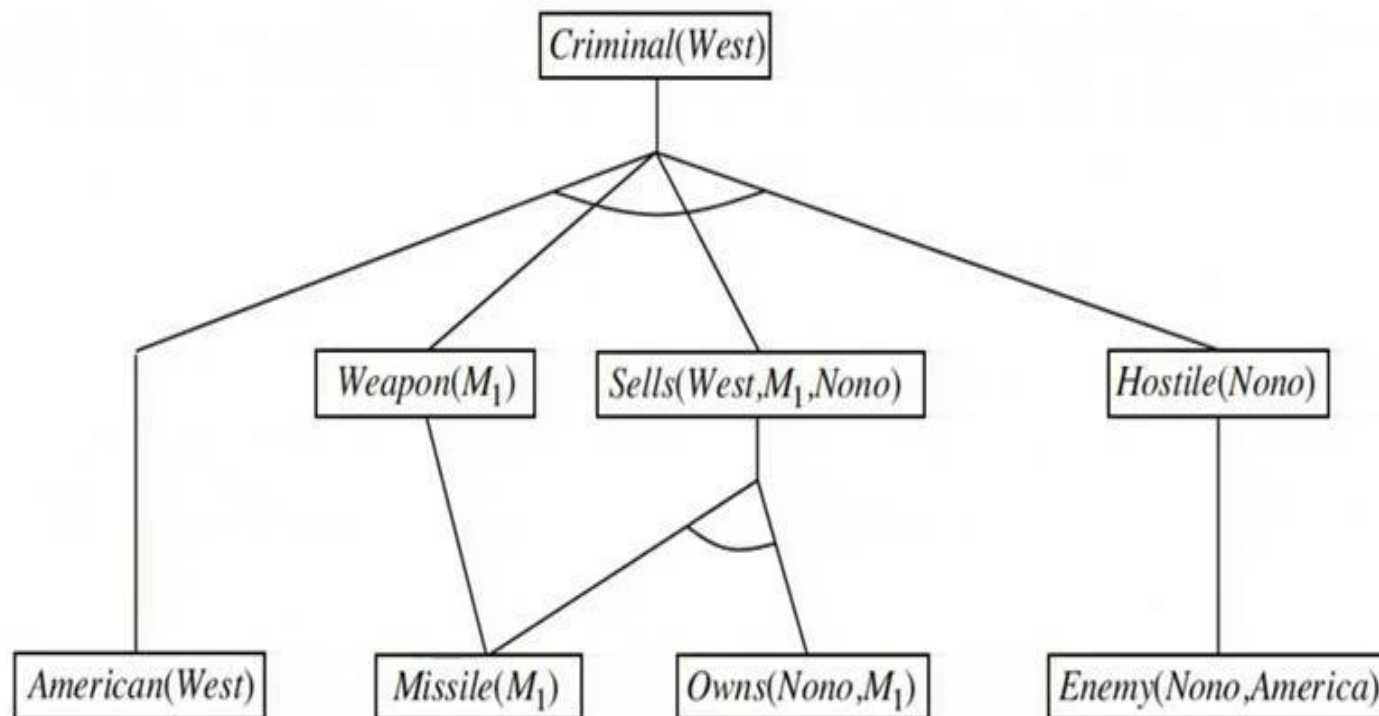
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Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Forward Chaining

Algorithm:

1. Start from the facts - Conjunct Ordering
2. Trigger all rules whose premises are satisfied - Pattern Matching
3. Add the conclusion to known facts – **Irrelevant Facts**
4. Repeat the steps till no new facts are added or the query is answered – Redundant Rule Matching

Backward Chaining

Algorithm:

1. Form Definite Clause
2. Start from the Goals
3. Search through rules to find the fact that support the proof
4. If it stops in the fact which is to be proved $\square$ Empty Set- LHS

Divide & Conquer Strategy

Substitution by Unification

Backward Chaining

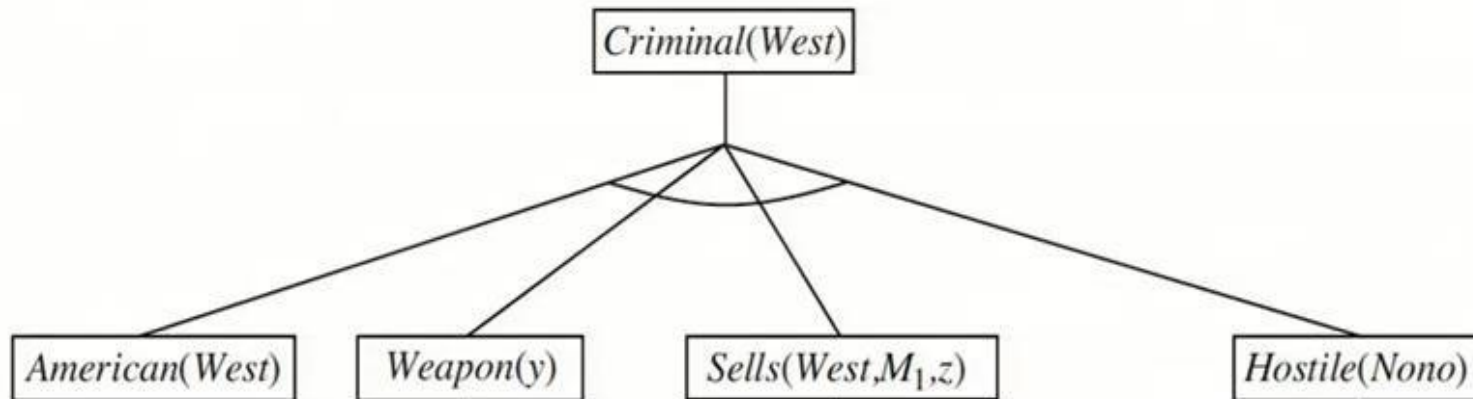
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- ③ $Missile(x) \Rightarrow Weapon(x)$
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Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Backward Chaining

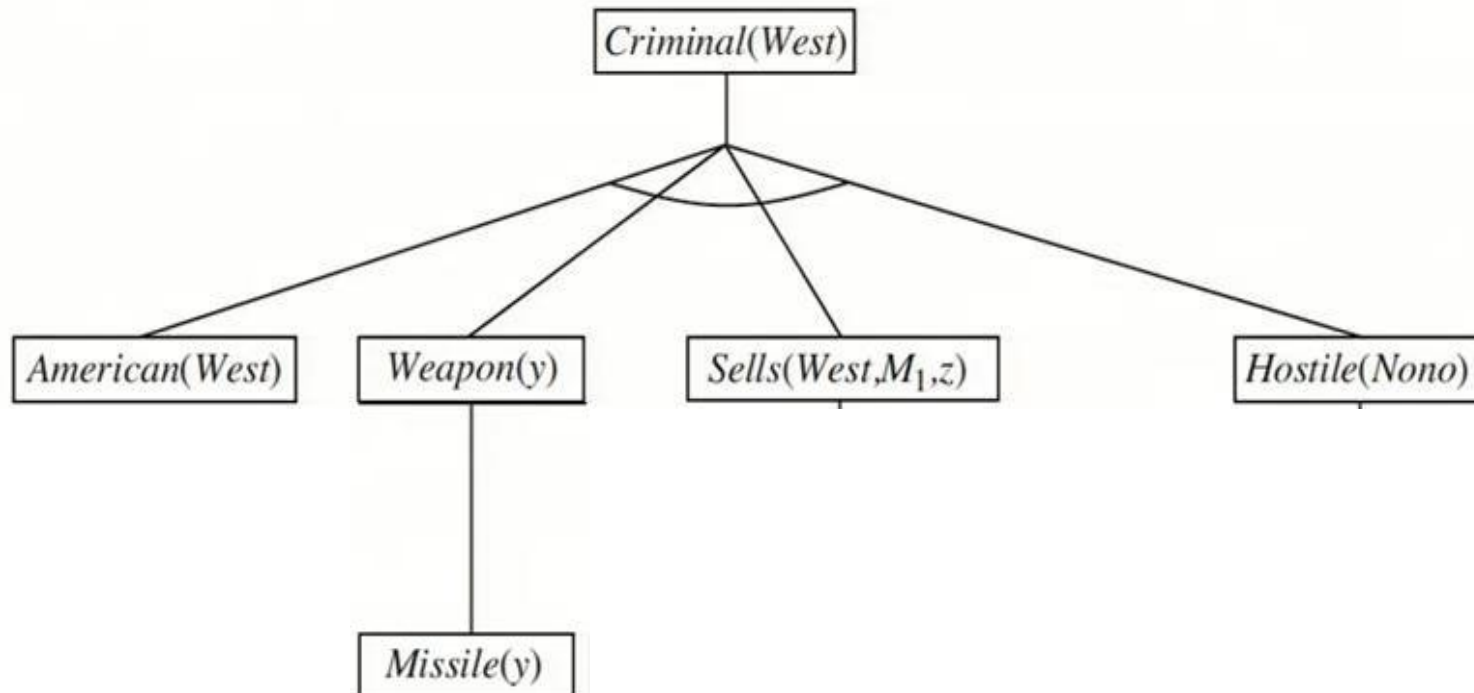
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- ③ $Missile(x) \Rightarrow Weapon(x)$
- ④ $Enemy(x, America) \Rightarrow Hostile(x)$

Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Backward Chaining

- ①

$American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \Rightarrow Criminal(x)$

②

$Missile(x) \wedge Owns(Nono, x) \Rightarrow Sells(West, x, Nono)$

③

$Missile(x) \Rightarrow Weapon(x)$

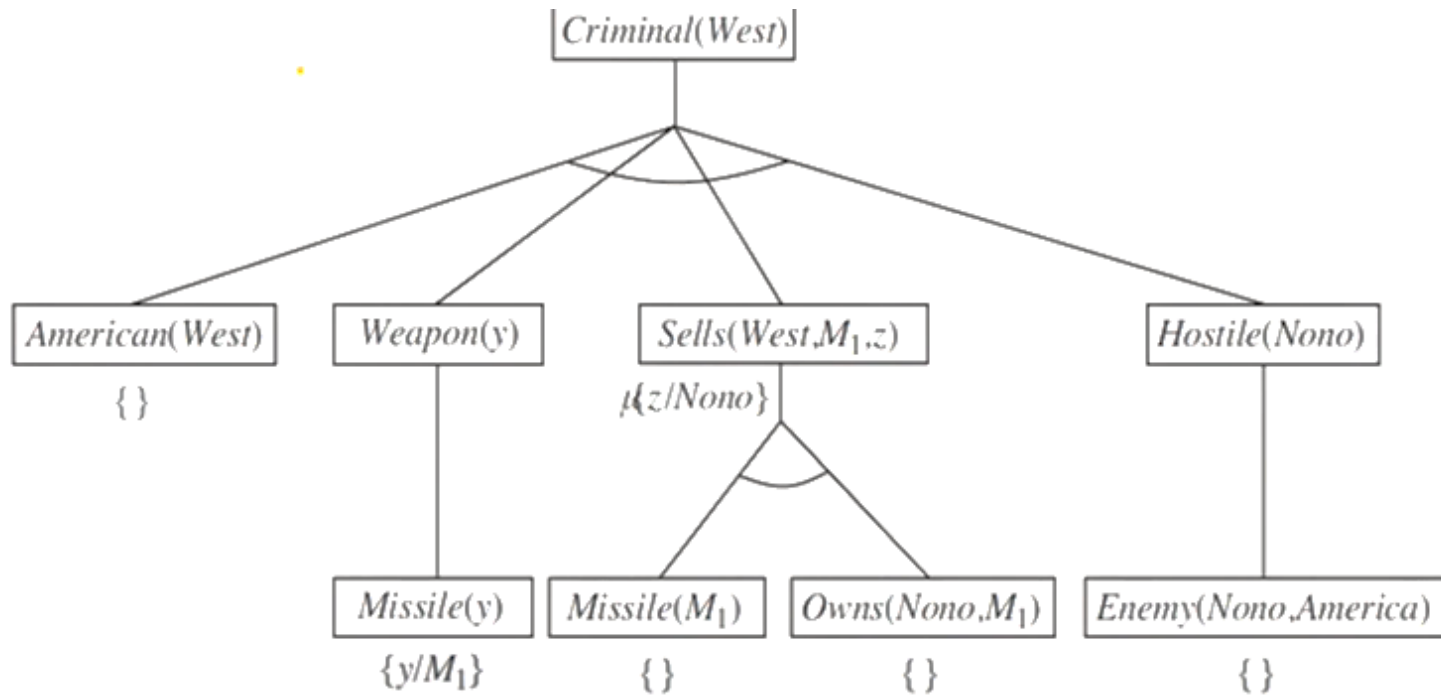
④

$Enemy(x, America) \Rightarrow Hostile(x)$
- Missile(M1)

Owns(Nono, M1)

American (West)

Enemy (Nono, America)



Exercise

Jack owns a dog. Every dog owner is an animal lover. No animal lover kills an animal. Either Jack or Curiosity killed the cat, who is named Tuna. Did Curiosity kill the cat?

These can be represented as follows:

- A. $(\exists x) \text{Dog}(x) \wedge \text{Owns}(\text{Jack}, x)$
- B. $(\forall x) ((\exists y) \text{Dog}(y) \wedge \text{Owns}(x, y)) \Rightarrow \text{AnimalLover}(x)$
- C. $(\forall x) \text{AnimalLover}(x) \wedge (\forall y) \text{Animal}(y) \Rightarrow \sim \text{Kills}(x, y)$
- D. $\text{Kills}(\text{Jack}, \text{Tuna}) \vee \text{Kills}(\text{Curiosity}, \text{Tuna})$
- E. $\text{Cat}(\text{Tuna})$
- F. $(\forall x) \text{Cat}(x) \Rightarrow \text{Animal}(x)$
- Q. $\text{Kills}(\text{Curiosity}, \text{Tuna})$

Citations and References



- Few content for these slides may have been obtained from prescribed books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs and gratefully acknowledge people others who made their course materials freely available online.
- .I have provided source information wherever necessary
- This is not a full fledged reading materials. Students are requested to refer to the textbook w.r.t detailed content of the presentation deck that is expected to be shared over e-learning portal - taxilla.
- I have added and modified the content to suit the requirements of the class dynamics & live session's lecture delivery flow for presentation
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- From BITS Pilani WILP: Prof.Raja vadhana, Prof. Indumathi, Prof.Sangeetha
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